Avi I. Flamholz Ph.D.

Assistant Professor & Head of Lab · The Rockefeller University

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Summary

I am excited when physical and mathematical principles help explain the behavior and evolution of cells, organisms and ecosystems. When possible, I strive to connect theory and laboratory experiments to real-world environments. I am focused on Earth's carbon cycle and working towards quantitative understanding of the metabolic rates of spatially-structured microbial communities in 3D environments like soils. Beyond understanding, we must also build technologies that erase our impacts on the Earth. I apply synthetic biology towards developing an engineered microbial agriculture with radically reduced land, water and emissions.

Education

University of California, Berkeley *Berkeley, CA*

PhD, Molecular and Cell Biology *2013 - 2019*

- Dissertation: "Analysis and Reconstitution of a Bacterial CO_2-Concentrating Mechanism"; Advisor: Prof. David Savage

Princeton University *Princeton, NJ*

B.S.E., Computer Science *2003 - 2007*

- Graduated *magna cum laude*, certificate in Applied & Computational Mathematics; Thesis advisor: Prof. Kai Li

Professional Experience

2025-	**Asst. Prof. & Head of The Laboratory of Environmental Microbiology**, The Rockefeller University
2020-2024	**Postdoctoral Scholar in Biology & Bioengineering**, Dianne Newman & Rob Phillips Groups, Caltech
2013-2019	**Graduate Student Researcher**, David Savage Group, UC Berkeley
2012-2013	**Founding Technical Lead**, Via Transportation, New York City
2010-2012	**Pre-doctoral Researcher**, Ron Milo Group, Weizmann Institute, Israel
2007-2010	**Software Engineer II**, Google, New York City

Awards & Honors

2025-2029	**Irma T. Hirschl Trust**, Career Scientist Award
2024-2029	**Burroughs Wellcome Fund**, Careers at the Scientific Interface Transitional Award
2023	**Caltech Center for Environmental Microbial Interactions**, CEMI Travel Award
2022	**Western Photosynthesis Annual Meeting**, Best Talk Award
2021-2024	**Jane Coffin Childs Memorial Foundation**, Postdoctoral Fellowship
2020	**Caltech Center for Environmental Microbial Interactions**, CEMI Pilot Award
2015	**UC Berkeley MCB Department**, Travel Award
2019	**American Chemical Society**, Editor's Choice Award (Flamholz et al. Biochem.)
2014-2019	**National Science Foundation**, Graduate Research Fellowship
2007	**Princeton University Department of Computer Science**, Department Service Award
2007	**Elected to Sigma Xi**, Research Honors Society

Publications

[*]equal contribution, [+]mentored undergraduate, [^]mentored graduate student

Jansson JK, **Flamholz AI**, Peixoto R, et al. Heterotrophic respiration by soil microbes in a changing climate. Nat. Rev. Earth Environ. 2026:1-17. doi:10.1038/s43017-026-00806-x

Branning JM Jr, Ruff BM, Saccomano SC, **Flamholz AI**, Newman DK, Cash KJ. A fluorous-phase oxygen optical nanosensor for mitigating redox-active microbial metabolite interference. Analyst. 2026. doi:10.1039/d6an00043f

Lovat S, Ben-Nissan R, Milshtein E, Leger D, **Flamholz AI**, Tzachor A, Noor E, Milo R. *Electro-microbial production techno-economic viability and environmental implications*. Nature Biotechnology. 2025; doi:10.1038/s41587-025-02632-w

Eshel G, **Flamholz AI**, Shepon A, Milo R. *US grass-fed beef is as carbon intensive as industrial beef and $\approx$10-fold more intensive than common protein-dense alternatives*. PNAS. 2025; 122 (12): e2404329122. doi:10.1073/pnas.2404329122

Prywes N, Phillips NR, Oltrogge LM, Lindner S, Tsai YCC, de Pins B, Cowan AE, Taylor-Kearney LJ, Chang HA, Hall LN, Bellieny-Rabelo D, HM Nisonoff, Weissman RF, **Flamholz AI**, Ding D, Bhat AY, Shih PM, Mueller-Cajar O, Milo R, Savage DF. *A map of the rubisco biochemical landscape*. Nature. 2025; 1-6. doi:10.1038/s41586-024-08455-0

Flamholz AI, Goyal A, Fischer WW, Newman DK, Phillips R. *The proteome is a terminal electron acceptor*. PNAS. 2025; 122 (1): e2404048121. doi:/10.1073/pnas.2404048121.

Flamholz AI, Goldford J, Richter PA, Larsson EM, Jinich A, Fischer WW, Newman DK. *Annotation-free prediction of microbial oxygen utilization.* mSystems. 2024; e0076324. doi:10.1128/msystems.00763-24.

Goyal A, **Flamholz AI**, Petroff A, Murugan A. *Closed ecosystems extract energy through self-organized nutrient cycles*. PNAS. 2024; 120: e2309387120. doi:10.1073/pnas.2309387120

Wang RZ[^], Nichols RJ, Liu AK, **Flamholz AI**, Artier J, Banda DM, Savage DF, Eiler JM, Shih PM, Fischer WW. *Carbon isotope fractionation by an ancestral rubisco suggests biological proxies for CO_2 through geologic time should be re-evaluated.* PNAS. 2023; 120: e2300466120. doi:10.1073/pnas.2300466120
Commentary by Sven Kranz in PNAS.

Flamholz AI, Goyal A. *Spotlight: Matching metabolic supply to demand optimizes microbial growth*. Trends Microbiol. 2023; 31, 769–771. doi:10.1016/j.tim.2023.06.003

Lammers NC, **Flamholz AI**, Garcia HG. *Competing constraints shape the nonequilibrium limits of cellular decision-making*. PNAS. 2023;120: e2211203120 doi:10.1073/pnas.2211203120

Flamholz AI[*], Dugan E[*+], Panich J, Desmarais JJ, Oltrogge LM, Fischer WW, Singer SW, Savage DF. *Trajectories for the evolution of bacterial CO_2-concentrating mechanisms*. PNAS. 2022;119: e2210539119. doi:10.1073/pnas.2210539119
Highlighted by Phys.org, Caltech News, & InnovativeGenomics.org.

Flamholz AI, Saccomano S, Cash K, Newman DK. *Optical O_2 Sensors Also Respond to Redox Active Molecules Commonly Secreted by Bacteria*. MBio. 2022; e0207622. doi:10.1128/mbio.02076-22

Chure G[*], Banks RA[*], **Flamholz AI**, Sarai NS, Kamb M, Lopez-Gomez I, Bar-On Y, Milo R, Phillips R. Anthroponumbers.org: A quantitative database of human impacts on Planet Earth. Patterns. 2022;3: 100552. doi:10.1016/j.patter.2022.100552
Highlighted in Wired, Phys.org, & ScienceDaily.

Noor E[*], **Flamholz AI**[*], Jayaraman V[*], Ross BL[*], Cohen Y, Patrick WM, Gruic-Sovulj I, Tawfik DS. *Uniform binding and negative catalysis at the origin of enzymes*. Protein Science. 2022;31: e4381. doi:10.1002/pro.4381
Special issue posthumously honoring Dan Salah Tawfik.

Goldford JE, George AB, **Flamholz AI**, Segrè D. *Protein cost minimization promotes the emergence of coenzyme redundancy*. PNAS. 2022;119: e2110787119. doi:10.1073/pnas.2110787119

Flamholz AI, Newman DK. *Microbial communities: The metabolic rate is the trait*. Current Biology. 2022;32: R215–R218. doi:10.1016/j.cub.2022.02.002

Beber ME, Gollub MG, Mozaffari D, Shebek KM, **Flamholz AI**, Milo R, Noor E. *eQuilibrator 3.0: a database solution for thermodynamic constant estimation*. Nucleic Acids Research. 2021;50: D603–D609. doi:10.1093/nar/gkab1106

Greenwald HD, Kennedy LC, Hinkle A, Whitney ON, Fan VB, Crits-Christoph A, Harris-Lovett S, **Flamholz AI**, Al-Shayeb B, Liao LD, Beyers M, Brown D, Chakrabarti AR, Dow J, Frost D, Koekemoer M, Lynch C, Sarkar P, White E, Kantor R, Nelson KL. *Tools for interpretation of wastewater SARS-CoV-2 temporal and spatial trends demonstrated with data collected in the San Francisco Bay Area*. Water Research X. 2021;12: 100111. doi:10.1016/j.wroa.2021.100111

Sender R, Bar-On YM, Gleizer S, Bernshtein B, **Flamholz AI**, Phillips R, Milo R. *The total number and mass of SARS-CoV-2 virions*. PNAS. 2021;118: e2024815118. doi:10.1073/pnas.2024815118

Crits-Christoph A, Kantor RS, Olm MR, Whitney ON, Al-Shayeb B, Lou YC, **Flamholz AI**, Kennedy LC, Greenwald H, Hinkle A, Hetzel J, Spitzer S, Koble J, Tan A, Hyde F, Schroth G, Kuersten S, Banfield JF, Nelson KL. *Genome sequencing of sewage detects regionally prevalent SARS-CoV-2 variants*. MBio. 2021;12: e02703–20. doi:10.1128/mBio.02703-20

Claassens NJ, Scarinci G, Fischer A, **Flamholz AI**, Newell W, Frielingsdorf S, Lenz O, Bar-Even A. *Phosphoglycolate salvage in a chemolithoautotroph using the Calvin cycle*. PNAS. 2020;117: 22452–22461. doi:10.1073/pnas.2012288117

Flamholz AI, Dugan E[+], Blikstad C, Gleizer S, Ben-Nissan R, Amram S, Antonovsky N, Ravishankar S, Noor E, Bar-Even A, Milo R, Savage DF. *Functional reconstitution of a bacterial CO_2 concentrating mechanism in Escherichia coli*. Elife. 2020;9: e59882.
Commentary by Franklin & Jonikas in ELife, highlighted in Nature, Faculty Opinions.

Davidi D, Shamshoum M, Guo Z, Bar-On YM, Prywes N, Oz A, Jablonska J, **Flamholz AI**, Wernick DG, Antonovsky N, De Pins B, Shachar L, Hochhauser D, Peleg Y, Albeck S, Sharon I, Mueller-Cajar O, Milo R. *Highly active rubiscos discovered by systematic interrogation of natural sequence diversity*. EMBO Journal. 2020;39: e104081. doi:10.15252/embj.2019104081

Flamholz AI, Shih PM. *Cell biology of photosynthesis over geologic time*. Current Biology. 2020;30: R490–R494. doi:10.1016/j.cub.2020.01.076

Bar-On YM, **Flamholz AI**, Phillips R, Milo R. *Science Forum: SARS-CoV-2 (COVID-19) by the numbers*. Elife. 2020;9: e57309.
Highlighted in Small Things Considered.

Desmarais JJ[^], **Flamholz AI**, Blikstad C, Dugan EJ[+], Laughlin TG, Oltrogge LM, Chen AW[+], Wetmore K, Diamond S, Wang JY, Savage DF. *DABs are inorganic carbon pumps found throughout prokaryotic phyla. Nature Microbiology*. 2019;4: 2204–2215. doi:10.1038/s41564-019-0520-8
Commentary by Price, Long & Forster in Nature Microbiology

Flamholz AI, Prywes N, Moran U, Davidi D, Bar-On YM, Oltrogge LM, Alves R, Savage DF, Milo R. *Revisiting trade-offs between Rubisco kinetic parameters*. Biochemistry. 2019;58: 3365–3376. doi:10.1021/acs.biochem.9b00237
Editor's Choice Award

Blikstad C, **Flamholz AI**, Oltrogge LM, Savage DF. *Learning to Build a β-Carboxysome*. Biochemistry. 2019. pp. 2091–2092.

Jinich A, **Flamholz AI**, Ren H, Kim S-J, Sanchez-Lengeling B, Cotton CAR, Noor E, Aspuru-Guzik A, Bar-Even A. *Quantum chemistry reveals thermodynamic principles of redox biochemistry*. PLoS Comput Biology. 2018;14: e1006471. doi:10.1371/journal.pcbi.1006471

Noor E, **Flamholz AI**, Bar-Even A, Davidi D, Milo R, Liebermeister W. *The protein cost of metabolic fluxes: prediction from enzymatic rate laws and cost minimization*. PLoS Comput Biology. 2016;12: e1005167. doi:10.1371/journal.pcbi.1005167

Mangan NM[*], **Flamholz AI[*]**, Hood RD, Milo R, Savage DF. *pH determines the energetic efficiency of the cyanobacterial CO_2 concentrating mechanism*. PNAS. 2016;113: E5354–E5362. doi:10.1073/pnas.1525145113

Hood RD, Higgins SA, **Flamholz AI**, Nichols RJ, Savage DF. *The stringent response regulates adaptation to darkness in the cyanobacterium Synechococcus elongatus*. PNAS. 2016;113: E4867–E4876. doi:10.1073/pnas.1524915113

Nadler DC, Morgan S-A, **Flamholz AI**, Kortright KE, Savage DF. *Rapid construction of metabolite biosensors using domain-insertion profiling*. Nature Communications. 2016;7: 12266. doi:10.1038/ncomms12266

Oakes BL, Nadler DC, **Flamholz AI**, Fellmann C, Staahl BT, Doudna JA, Savage DF. *Profiling of engineering hotspots identifies an allosteric CRISPR-Cas9 switch*. Nature Biotechnology. 2016;34: 646–651. doi:10.1038/nbt.3528

Davidi D, Noor E, Liebermeister W, Bar-Even A, **Flamholz AI**, Tummler K, Barenholz U, Goldenfeld M, Shlomi T, Milo R. *Global characterization of in vivo enzyme catalytic rates and their correspondence to in vitro k_{cat} measurements*. PNAS. 2016;113: 3401–3406.

Flamholz AI, Phillips R, Milo R. *The quantified cell*. Molecular Biology of the Cell. 2014;25: 3497–3500. doi:10.1091/mbc.E14-09-1347

Liebermeister W, Noor E, **Flamholz AI**, Davidi D, Bernhardt J, Milo R. *Visual account of protein investment in cellular functions*. PNAS. 2014;111: 8488–8493. doi:10.1073/pnas.1314810111

Noor E, Bar-Even A, **Flamholz AI**, Reznik E, Liebermeister W, Milo R. *Pathway thermodynamics highlights kinetic obstacles in central metabolism*. PLoS Computational Biology. 2014;10: e1003483. doi:10.1371/journal.pcbi.1003483

Noor E, **Flamholz AI**, Liebermeister W, Bar-Even A, Milo R. *A note on the kinetics of enzyme action: a decomposition that highlights thermodynamic effects*. FEBS Letters. 2013;587: 2772–2777. doi:10.1016/j.febslet.2013.07.028

Flamholz AI[*], Noor E[*], Bar-Even A, Liebermeister W, Milo R. *Glycolytic strategy as a tradeoff between energy yield and protein cost*. PNAS. 2013;110: 10039–10044. doi:10.1073/pnas.1215283110
Commentary by Stettner & Segre in PNAS

Zelcbuch L, Antonovsky N, Bar-Even A, Levin-Karp A, Barenholz U, Dayagi M, Liebermeister W, **Flamholz AI**, Noor E, Amram S, Brandis A, Bareia T, Yofe I, Jubran H, Milo R. *Spanning high-dimensional expression space using ribosome-binding site combinatorics*. Nucleic Acids Research. 2013;41: e98–e98. doi:10.1093/nar/gkt151

Bar-Even A, **Flamholz AI**, Noor E, Milo R. *Thermodynamic constraints shape the structure of carbon fixation pathways*. Biochimica et Biophysica Acta (BBA)-Bioenergetics. 2012;1817: 1646–1659. doi:10.1016/j.bbabio.2012.05.002

Noor E, Bar-Even A, **Flamholz AI**, Lubling Y, Davidi D, Milo R. *An integrated open framework for thermodynamics of reactions that combines accuracy and coverage*. Bioinformatics. 2012;28: 2037–2044. doi:10.1093/bioinformatics/bts317

Bar-Even A, Noor E, **Flamholz AI**, Milo R. *Design and analysis of metabolic pathways supporting formatotrophic growth for electricity-dependent cultivation of microbes*. Biochimica et Biophysica Acta (BBA)-Bioenergetics. 2012;1827: 1039–1047. doi:10.1016/j.bbabio.2012.10.013

Flamholz AI, Noor E, Bar-Even A, Milo R. *eQuilibrator—the biochemical thermodynamics calculator*. Nucleic Acids Research. 2012;40: D770–D775.

Bar-Even A, **Flamholz AI**, Noor E, Milo R. *Rethinking glycolysis: on the biochemical logic of metabolic pathways*. Nature Chemical Biology. 2012;8: 509–517.

Bar-Even A, **Flamholz AI**, Noor E, Milo R. *Thermodynamic constraints shape the structure of carbon fixation pathways*. Bioenergetics. 2012.

Bar-Even A, Noor E, **Flamholz AI**, Buescher JM, Milo R. *Hydrophobicity and charge shape cellular metabolite concentrations*. PLoS Comput Biology. 2011;7: e1002166. doi:10.1371/journal.pcbi.1002166

Huttenhower C, **Flamholz AI**, Landis JN, Sahi S, Myers CL, Olszewski KL, Hibbs MA, Siemers NO, Troyanskaya OG, Coller HA. *Nearest Neighbor Networks: clustering expression data based on gene neighborhoods*. BMC Bioinformatics. 2007;8: 1–13. doi:10.1186/1471-2105-8-250

Funded Research Proposals

June 2025. *Towards a microbial platform for sustainable production*. **Rockefeller SNF Institute** Idea Grant.

January 2025. *Understanding and engineering microbial contributions to the global carbon cycle*. **Irma T. Hirschl Trust** Careers Scientist Award.

June 2023. *Predicting microbial CO_2 production in global soils*. **Burroughs Wellcome Fund** Careers at the Scientific Interface Transitional Fellowship.

February 2023. *Predicting the Fate of Carbon in Soils using Language Models on Sequence Data from Microbial Communities*. **Schmidt Academy for Software Engineering.** With W. Fischer; supporting Philippa Richter for 2 years.

July 2021. *Developing bacterial biofilms as a model for predicting tissue metabolism*. **Jane Coffin Childs Memorial Foundation** Postdoctoral Fellowship.

September 2020. *Experimental and theoretical approaches to studying the physiology of single cells in structured environments*. **Caltech Center for Emerging Microbial Interactions** Pilot Grant.

May 2020. *Early detection of COVID-19 reemergence via wastewater surveillance of SARS-CoV-2* written with Profs. Kara Nelson & Jill Banfield's groups. **Innovative Genomics Institute at UC Berkeley** Rapid Response COVID-19 Research Grant.

September 2018. *Mapping sequence-function landscapes to isolate improved Rubiscos* with Prof. David Savage. **National Science Foundation** award #1818377

September 2013. *Interrogating Overflow Metabolism with Laboratory Evolution* **National Science Foundation** Graduate Research Fellowship

Presentations

Select Talks

08/2026 - Function of Evolving Systems GRC, Waterville Valley, NH

05/2026 - Coarse-Graining Microbial Communities, Kavli Institute for Theoretical Physics at UCSB, Santa Barbara, CA

05/2026 - Princeton/CUNY Physics of Life along the Northeast Corridor Symposium, CUNY Graduate Center, New York, NY

03/2026 - Climate & Ecology Session, Annual Meeting of the American Physical Society, Denver, CO

03/2026 - Physical Biology Course, Cold Spring Harbor, NY

12/2025 - Terrestrial Ecosystems Session, Annual Meeting of the American Geophysical Union, New Orleans, LA

12/2025 - Biology & Paleo-Environment Seminar, Lamont Doherty Earth Observatory of Columbia University, Palisades, NY

11/2025 - Weekly Seminar, Cary Institute of Ecosystem Studies, Millbrook, NY

10/2025 - National Institute for Theory and Mathematics in Biology, Microbial Communities Workshop, Chicago, IL

10/2025 - Memorial Sloan Kettering, Computational Biology & Medicine Seminar, New York, NY

07/2025 - Metabolic Pathway Analysis Conference, University of Vienna, Vienna, Austria

04/2025 - Ohio State University Department of Microbiology, Columbus, OH

03/2025 - ENIGMA Group at Rutgers University, New Brunswick, NJ

03/2025 - Yale University Department of Molecular, Cellular and Developmental Biology, New Haven, CT

03/2025 - Economic Principles in Cell Physiology online forum

11/2024 - The Salk Institute Theory Seminar, San Diego, CA

11/2024 - USC Systems Biology Graduate Course, Los Angeles, CA

11/2024 - USC Department of Marine and Environmental Sciences, Los Angeles, CA

10/2024 - Joint BioEnergy Institute, Emeryville, CA

5/2024 - Caltech Center for Evolutionary Science, Pasadena, CA

4/2024 - Southern CA Annual Geobiology Meeting, Pasadena, CA

04/2024 - MIT Department of Earth and Planetary Sciences, Cambridge, MA

03/2024 - The Rockefeller University, New York, NY

03/2024 - American Physical Society March Meeting, Quantitative Physiology Section, Minneapolis, MN

02/2024 - Stanford University Department of Bioengineering, Stanford, CA

02/2024 - Harvard Medical School Department of Systems Biology, Boston, MA

02/2024 - The Rockefeller University, New York, NY

11/2023 - Caltech Division of Geology and Planetary Sciences, Pasadena, CA

03/2023 - MIT Department of Earth and Planetary Sciences, Cambridge, MA

07/2022 - CCM10 Conference, Princeton University, Princeton, NJ

03/2022 - Western Photosynthesis Conference (virtual). *Awarded "best talk."*

07/2021 - Microbial Ecology and Evolution Course, Kavli Institute for Theoretical Physics, Goleta, CA

08/2020 - Australian National University Department of Plant Biology (virtual)

07/2020 - York University Department of Plant Biology (virtual)

07/2018 - International Geobiology Course, Catalina Island, CA

05/2017 - Amyris Inc, Emeryville, CA

11/2013 - UCSF Theory Lunch, San Francisco, CA

09/2013 - Metabolic Pathway Analysis Conference, Oxford University, Oxford, UK

Teaching Experience

Spring '26	**Biology by the Numbers**,		*Rockefeller U.*
Fall '24	**The history and future of atmospheric dioxygen**, Co-teaching with Woody Fischer		*Caltech*
Sum. '19	**The Microverse**, Introduction to microscopy & programming, Clubes de Ciencia		*Monterrey, MX*
Spring '16	**Biochemistry: Pathways, Mechanisms, and Regulation**, Graduate Student Instructor		*UC Berkeley*
Sum. '15	**QB3 Python Programming Bootcamp**, Teaching Assistant		*UC Berkeley*
Fall '15	**General Microbiology**, Graduate Student Instructor		*UC Berkeley*
Fall '11	**English as a Second Language**, Mesila program for refugees		*Tel Aviv, Israel*
Fall '09	**Introduction to Programming, 5th-8th Grades**, Citizen Schools Volunteer		*Brooklyn, NY*
2005-2007	**Head laboratory teaching assistant for intro courses**, Dept. of Computer Science		*Princeton*

Formal Mentorship

Postdocs

2026- **Dr. Julie McDonald**, Postdoctoral Fellow *Rockefeller U.*

Graduate Students

2025- **James Wang**, NSF Graduate Research Fellow *Rockefeller U.*

Post-Baccalaureate

2025- **Eden Elkayam**, Research Assistant *Rockefeller U.*
2025- **Sophia Adami-Sampson**, Research Assistant *Rockefeller U.*
2022-2025 **Philippa Richter**, Schmidt Scholar in the Fischer Lab - *NDSEG PhD fellow at UC Berkeley* *Caltech*
2015-2020 **Eli Dugan**, Technician in Savage Lab - *PhD student at UCSF* *UC Berkeley*

Summer and Visiting Students

2026 **Alexandra Durt**, Chemers Neustein SURF Summer Student *Rockefeller U.*
2026 **Zainab Aleem**, Bard-Rockefeller Semester in Science Program *Rockefeller U.*
2025 **Albert Li**, Chemers Neustein SURF Summer Student *Rockefeller U.*
2022 **Gia Han Vuong**, Caltech CEMI-WAVE Summer Research Fellow in the Newman Lab *Caltech*
2019 **Alejandra Zapata**, UCB Transfers to Excellence Summer Research Fellow in the Savage Lab *UC Berkeley*

Rotation Students

2026 **Nina Magid**, David Rockefeller Graduate Program *Rockefeller U.*
2026 **Julie Chen**, David Rockefeller Graduate Program *Rockefeller U.*
2021 **Elin Larsson**, Rotation student in the Newman Lab - *joined for PhD* *Caltech*
2017 **Edward Koleski**, Rotation student in the Savage Lab *UC Berkeley*
2017 **Julia Borden**, Rotation student in the Savage Lab - *joined for PhD* *UC Berkeley*
2015 **Dylan McLung**, Rotation student in the Savage Lab *UC Berkeley*

Undergraduate Researchers

2016-2018 **Allen Chen**, Savage Lab Undergraduate Researcher - *now a PhD student at Caltech* *UC Berkeley*
2015-2016 **Sumedha Ravishankar**, Savage Lab Undergrad Researcher - *now a postdoc at UCSD* *UC Berkeley*

Service and Outreach

2011-Present. *Maintain and develop eQuilibrator - the biochemical thermodynamics calculator*. With Elad Noor; 500-1000 students and researchers use eQuilibrator each month. `equilibrator.weizmann.ac.il`

Summer 2022. *Caltech PCC CEMI-WAVE summer research mentor*. Volunteer mentor for a summer program bringing Pasadena Community College students to Caltech for paid summer research.

Summer 2019. *Clubes de Ciencias, Monterrey, Mexico*. Volunteer teacher. Designed and co-taught "The Microverse" - a one-week course on microscopy and introductory programming for 20 students. With Alejandro Balderas of Tecnológico de Monterrey.

Summer 2019. *UC Berkeley Transfers to Excellence*. Volunteer mentor for a summer program bringing prospective transfer students to UC Berkeley for paid research internships.

2018. *Student representative to Graduate Admissions Committee*. UC Berkeley Department of Molecular and Cell Biology.

2017. *Student representative to "host-microbe interactions" Faculty Search Committee*. UC Berkeley Department of Molecular and Cell Biology.

2017-2019. *Primary organizer of annual 100-person campus-wide symposium "Photosynthesis, Carbon-Fixation and the Environment."* UC Berkeley.

2017. *Student organizer of Biophysics, Biochemistry and Structural Biology divisional retreat*. UC Berkeley Department of Molecular and Cell Biology.

2013-2017. *Primary organizer of UC Berkeley Systems Biology Reading Group*.

Peer Review

I have refereed papers for *ACS Synthetic Biology, Biogeosciences, Biophysical Journal, BioEssays, Cell, Current Opinion in Chemical Biology, ELife, Free Radical Biology and Medicine, mSystems, Nature, PLoS One, PNAS, PRX Life* and *Science*.

Mentors

B.S.E Junior Research Paper **Prof. Hilary Coller**, Princeton University, Department of Molecular Biology (now UCLA)

B.S.E. Senior Thesis **Prof. Kai Li**, Princeton University, Department of Computer Science

Pre-doctoral Research Mentor **Prof. Ron Milo**, Weizmann Institute for Science, Department of Plant and Environmental Sciences

PhD Mentor **Prof. David Savage**, University of California, Berkeley, Department of Molecular and Cell Biology

Postdoc Co-mentors at Caltech
Prof. Dianne Newman Divisions of Biology & Biological Engineering, Geological and Planetary Science
Prof. Rob Phillips Department of Physics, Division of Biology and Biological Engineering
Woodward Fischer, Professor of Geobiology, California Institute of Technology